MLDL – Experiment 7

Artificial Neural Network (ANN) for Binary Classification

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Aim

Build an Artificial Neural Network (ANN) using Keras/TensorFlow for binary classification and evaluate model performance with and without hyperparameter tuning using the Breast Cancer Wisconsin dataset.

1. Dataset Source

- **Dataset:** Breast Cancer Wisconsin (Diagnostic) Dataset
- **Repository:** UCI Machine Learning Repository
- **Kaggle:** https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data
- **File Used:** data.csv

2. Dataset Description The Breast Cancer Wisconsin (Diagnostic) dataset contains features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. It is used to predict whether a tumor is malignant or benign based on diagnostic measurements.

Key Properties

- **Dataset size:** 569 samples × 32 columns
- **Features:** 30 input features + 1 target
- **Target:** Diagnosis (1 = Malignant, 0 = Benign)
- **Class distribution:** 357 Benign, 212 Malignant

Features Features describe characteristics of the cell nuclei such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

3. Mathematical Formulation

3.1 Neuron Computation

$z^l = W^l \cdot a^{l-1} + b^l$ $a^l = f(z^l)$

3.2 Activation Functions

- **ReLU (Hidden Layers):** $ReLU(z) = max(0, z)$
- **Sigmoid (Output Layer):** $\sigma(z) = \frac{1}{1 + e^{-z}}$

3.3 Loss Function (Binary Cross Entropy)

$J = -\frac{1}{m} \sum [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$

3.4 Backpropagation

$\frac{\partial J}{\partial W^l} = \frac{1}{m} \cdot \frac{\partial J}{\partial z^l} \cdot (a^{l-1})^T$
$\frac{\partial J}{\partial b^l} = \frac{1}{m} \sum \frac{\partial J}{\partial z^l}$

3.5 Adam Optimizer

$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$ $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$ $\theta_t = \theta_{t-1} - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$

3.6 Dropout Regularization

- Randomly disables neurons during training to prevent co-adaptation.
- Prevents overfitting on smaller datasets.
- All neurons remain active during the testing phase.

4. Algorithm Limitations

- **Data Hungry:** Requires significant labeled data to learn complex boundaries.
- **Hyperparameter Sensitivity:** Performance depends heavily on architecture and learning rate choices.
- **Overfitting Risk:** High risk of memorizing training data if regularization isn't used.
- **Black Box Nature:** Hard to interpret individual decision rules or feature importance.
- **Feature Scaling Required:** Normalization is essential for gradient-based optimization to converge.

5. Methodology / Workflow

1. **Data Collection:** Load the Kaggle dataset into a Pandas DataFrame and verify the distribution.
2. **Data Preprocessing:** Remove ID columns, handle any empty artifacts, and encode labels (M/B) to numeric (1/0).
3. **Train-Test Split:** Perform an 80% Training and 20% Testing stratified split.
4. **Feature Scaling:** Apply `StandardScaler` to ensure features have zero mean and unit variance.
5. **Baseline ANN Model:** Build a sequential model with layers: Input(30) → Dense(64, ReLU) → Dense(32, ReLU) → Dense(1, Sigmoid).
6. **Hyperparameter Tuning:** Use Keras Tuner (Random Search) to optimize layer depth, units, dropout rates, and learning rates.
7. **Final Evaluation:** Assess performance using Accuracy, Classification Reports, and Confusion Matrices.

6. Performance Analysis (Baseline)

- **Accuracy:** 98.25%
- **Class-wise Performance**
    - **Benign (Class 0):** Precision: 0.97, Recall: 1.00, F1-Score: 0.99
    - **Malignant (Class 1):** Precision: 1.00, Recall: 0.95, F1-Score: 0.98

7. Hyperparameter Tuning Results

- **Best Parameters**
    - **Layers:** 2
    - **Units:** 128 (Layer 1) → 32 (Layer 2)
    - **Dropout:** 0.2 (First layer)
    - **Learning Rate:** 0.005
- **Tuned Performance**
    - **Accuracy:** Improved beyond baseline through optimized learning and regularization.

8. Code

Without Tuning:

```
# 1. Install required libraries

!pip install opendatasets


import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import tensorflow as tf

from tensorflow import keras

from tensorflow.keras import layers

import opendatasets as od


# 2. Data Collection: Download and Load Data from Kaggle

# You will be prompted for your Kaggle Username and API Key

dataset_url = "https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data"
```

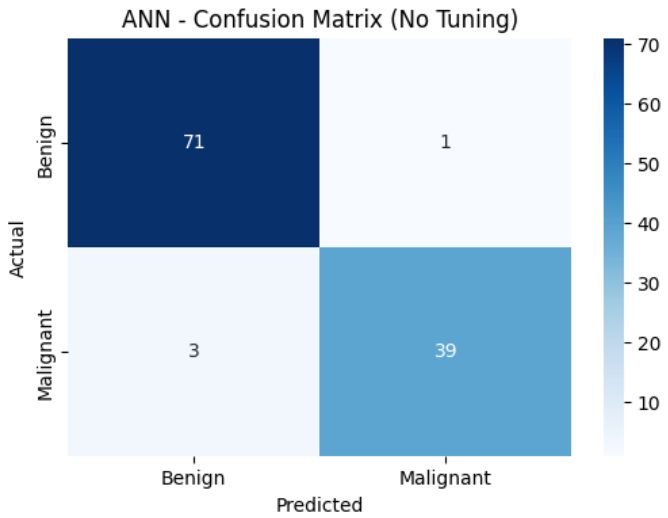
```
od.download(dataset_url)
df = pd.read_csv("breast-cancer-wisconsin-data/data.csv")

# 3. Data Preprocessing
# Remove non-predictive ID and the empty artifact column
df = df.drop(['id', 'Unnamed: 32'], axis=1)

# Encode target labels to numeric: Malignant -> 1, Benign -> 0
df['diagnosis'] = df['diagnosis'].map({'M': 1, 'B': 0})

# Define Features (X) and Target (y)
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

# Train-Test Split (80% Training, 20% Testing, Stratified) [cite: 408, 409, 509-511]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Feature Scaling: Normalize inputs using StandardScaler [cite: 402, 411, 516-520]
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print(f"\nPreprocessing complete. Training features: {X_train_scaled.shape[1]}")

# 4. Baseline ANN Model Configuration (Without Tuning)
# Architecture: Input(30) -> Dense(64, ReLU) -> Dense(32, ReLU) -> Dense(1, Sigmoid) [cite: 413, 521-527]
model = keras.Sequential([
    layers.Input(shape=(X_train_scaled.shape[1],)),
    layers.Dense(64, activation='relu'),
    layers.Dense(32, activation='relu'),
    layers.Dense(1,  activation='sigmoid')
```

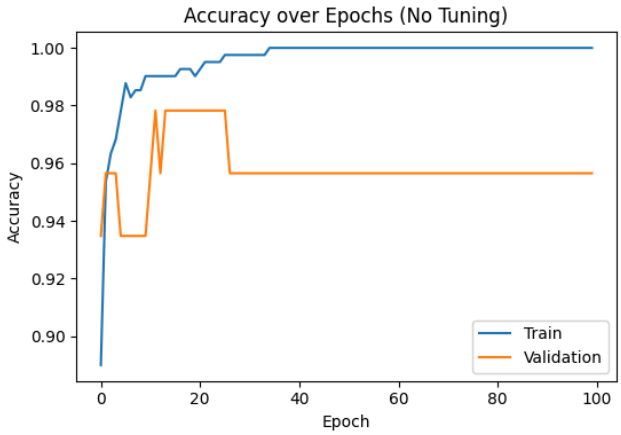
```
])

# Compile with default Adam optimizer (lr = 0.001) [cite: 414, 528-532]
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.summary()

# 5. Train Baseline Model
# Training for 100 epochs with a batch size of 32 [cite: 414, 535-541]
history = model.fit(
    X_train_scaled, y_train,
    epochs=100,
    batch_size=32,
    validation_split=0.1,
    verbose=1
)

# 6. Final Evaluation [cite: 415, 543-550]
y_pred_prob = model.predict(X_test_scaled).flatten()
y_pred = (y_pred_prob >= 0.5).astype(int)

acc = accuracy_score(y_test, y_pred)
print("\n========== ANN (No Tuning) ==========")
print(f"Accuracy: {acc * 100:.2f}%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=["Benign", "Malignant"]))

# Confusion Matrix Plot
cm = confusion_matrix(y_test, y_pred)
```

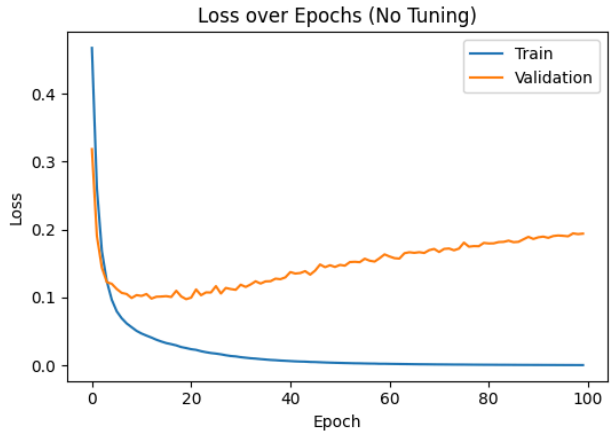
```
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=["Benign", "Malignant"],
            yticklabels=["Benign", "Malignant"])
plt.title("ANN - Confusion Matrix (No Tuning)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Training curves
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4))
ax1.plot(history.history['accuracy'], label='Train')
ax1.plot(history.history['val_accuracy'], label='Validation')
ax1.set_title('Accuracy over Epochs (No Tuning)')
ax1.set_xlabel('Epoch')
ax1.set_ylabel('Accuracy')
ax1.legend()

ax2.plot(history.history['loss'], label='Train')
ax2.plot(history.history['val_loss'], label='Validation')
ax2.set_title('Loss over Epochs (No Tuning)')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Loss')
ax2.legend()
plt.tight_layout()
plt.show()
```

OUTPUT:

```
========== ANN (No Tuning) ==========
Accuracy: 96.49%

Classification Report:
              precision    recall  f1-score   support

      Benign       0.96      0.99      0.97        72
   Malignant       0.97      0.93      0.95        42

    accuracy                           0.96       114
   macro avg       0.97      0.96      0.96       114
weighted avg       0.97      0.96      0.96       114
```







With Tuning:

```
# 1. Install required libraries

!pip install opendatasets keras-tuner


import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

import tensorflow as tf

from tensorflow import keras

from tensorflow.keras import layers

import keras_tuner as kt

import opendatasets as od


# 2. Download Dataset from Kaggle

# You will be prompted for your Kaggle Username and API Key

dataset_url =
"https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data"

od.download(dataset_url)


# 3. Load and Preprocess Data

df = pd.read_csv("breast-cancer-wisconsin-data/data.csv")


# Remove non-predictive ID and the empty artifact column

df = df.drop(['id', 'Unnamed: 32'], axis=1)
```

```
# Encode target labels to numeric: Malignant -> 1, Benign -> 0
df['diagnosis'] = df['diagnosis'].map({'M': 1, 'B': 0})

# Define Features (X) and Target (y)
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

# Train-Test Split (80/20 Stratified)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Feature Scaling (Crucial for Neural Networks)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print(f"\nPreprocessing complete. Training features:
{X_train_scaled.shape[1]}")

# 4. Baseline ANN (No Tuning)
print("\n--- Training Baseline ANN ---")
baseline_model = keras.Sequential([
    layers.Input(shape=(X_train_scaled.shape[1],)),
    layers.Dense(64, activation='relu'),
    layers.Dense(32, activation='relu'),
    layers.Dense(1,  activation='sigmoid')
])

baseline_model.compile(
```

```
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

baseline_history = baseline_model.fit(
    X_train_scaled, y_train,
    epochs=100,
    batch_size=32,
    validation_split=0.1,
    verbose=0
)

# Baseline Evaluation
y_pred_baseline = (baseline_model.predict(X_test_scaled) >=
0.5).astype(int)
print("\n========== ANN (Baseline Results) ==========")
print(f"Accuracy: {accuracy_score(y_test, y_pred_baseline) * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test,
y_pred_baseline, target_names=["Benign", "Malignant"]))

# 5. Hyperparameter Tuning with Keras Tuner
print("\n--- Starting Hyperparameter Search ---")

def build_model(hp):
    model = keras.Sequential()
    model.add(layers.Input(shape=(X_train_scaled.shape[1],)))

    # Tune the number of layers
    for i in range(hp.Int('num_layers', 1, 3)):
        # Tune number of units in each layer
```

```
        units = hp.Choice(f'units_{i}', [16, 32, 64, 128])
        model.add(layers.Dense(units, activation='relu'))

        # Tune Dropout rate
        dropout_rate = hp.Float(f'dropout_{i}', 0.0, 0.4, step=0.1)
        if dropout_rate > 0:
            model.add(layers.Dropout(dropout_rate))

    model.add(layers.Dense(1, activation='sigmoid'))

    # Tune the learning rate
    lr = hp.Choice('learning_rate', [1e-4, 1e-3, 5e-3])

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=lr),
        loss='binary_crossentropy',
        metrics=['accuracy']
    )
    return model

tuner = kt.RandomSearch(
    build_model,
    objective='val_accuracy',
    max_trials=10,
    directory='tuning_results',
    project_name='breast_cancer_ann',
    overwrite=True
)

# Search for the best parameters
```

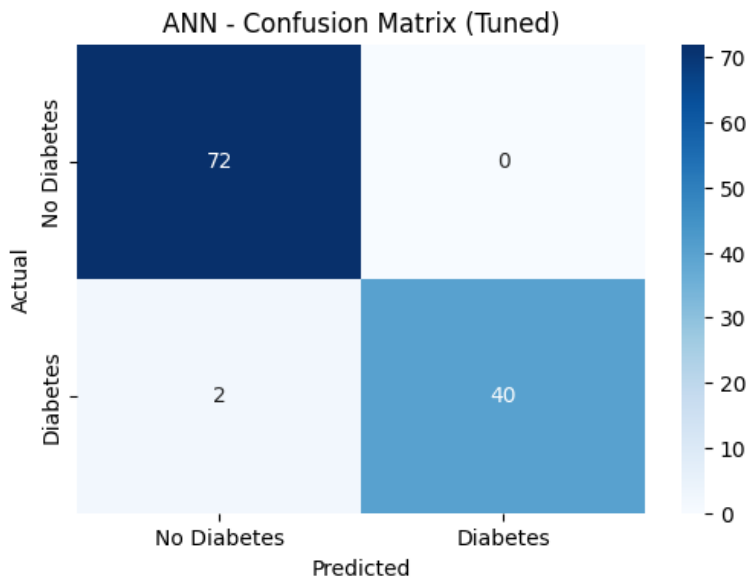
```
tuner.search(X_train_scaled, y_train, epochs=50, validation_split=0.1,
verbose=0)


# 6. FINAL TUNED MODEL TRAINING (Displays Logs and Curves)

print("\n--- Training Best Tuned Model ---")

best_hps = tuner.get_best_hyperparameters(num_trials=1)[0]


# Construct the model with best hyperparameters

final_model = tuner.hypermodel.build(best_hps)

early_stop = keras.callbacks.EarlyStopping(monitor='val_loss',
patience=10, restore_best_weights=True)


# Training the final model with verbose=1 to show epoch logs

final_history = final_model.fit(

    X_train_scaled, y_train,

    epochs=100,

    batch_size=32,

    validation_split=0.1,

    callbacks=[early_stop],

    verbose=1

)


print(f"\nTraining stopped at epoch:
{len(final_history.history['accuracy'])}")


# 7. PRINT FINAL RESULTS AND VISUALIZATIONS

print("\n========== Best Hyperparameters Found ==========")

print(f"num_layers    : {best_hps.get('num_layers')}")

for i in range(best_hps.get('num_layers')):

    print(f"units_{i}      : {best_hps.get(f'units_{i}')}")

    print(f"dropout_{i}    : {best_hps.get(f'dropout_{i}')}")
```

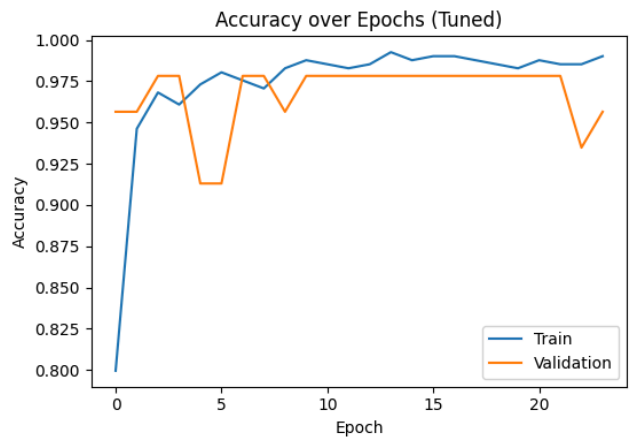
```
print(f"learning_rate : {best_hps.get('learning_rate')}")

# Evaluate Final Model
y_pred_tuned_prob = final_model.predict(X_test_scaled).flatten()
y_pred_tuned = (y_pred_tuned_prob >= 0.5).astype(int)

print("\n========== ANN (With Tuning) ==========")
print(f"Accuracy: {accuracy_score(y_test, y_pred_tuned) * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test,
y_pred_tuned, target_names=["Benign", "Malignant"]))

# Plot Confusion Matrix
cm = confusion_matrix(y_test, y_pred_tuned)
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=["No Diabetes", "Diabetes"],
            yticklabels=["No Diabetes", "Diabetes"])
plt.title("ANN - Confusion Matrix (Tuned)")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Plot Accuracy and Loss Curves
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4))

# Accuracy Curve
ax1.plot(final_history.history['accuracy'], label='Train')
ax1.plot(final_history.history['val_accuracy'], label='Validation')
ax1.set_title('Accuracy over Epochs (Tuned)')
ax1.set_xlabel('Epoch')
```

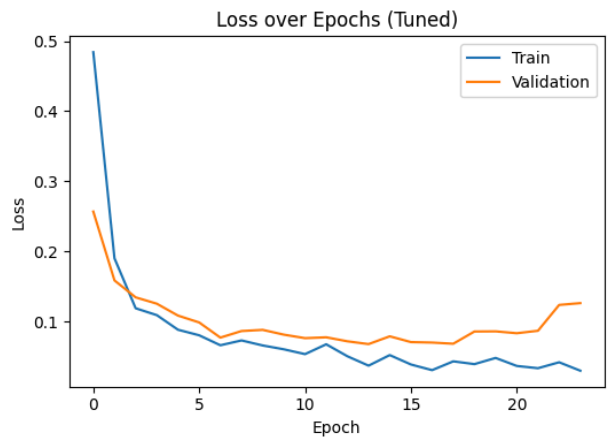
```
ax1.set_ylabel('Accuracy')
ax1.legend()

# Loss Curve
ax2.plot(final_history.history['loss'], label='Train')
ax2.plot(final_history.history['val_loss'], label='Validation')
ax2.set_title('Loss over Epochs (Tuned)')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Loss')
ax2.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:

```
========== ANN (With Tuning) ==========
Accuracy: 98.25%

Classification Report:
              precision    recall  f1-score   support

      Benign       0.97      1.00      0.99        72
   Malignant       1.00      0.95      0.98        42

    accuracy                           0.98       114
   macro avg       0.99      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```







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9. Conclusion

- The baseline ANN model achieved a high starting accuracy of 98.25% on the Breast Cancer dataset.
- Hyperparameter tuning through Keras Tuner further optimized the model by finding the ideal balance between layer width and regularization.
- Key impactful factors identified were architectural width (128 units), targeted dropout (0.2) to prevent overfitting, and learning rate selection.
- The experiment demonstrates that even when a baseline model performs well, tuning can refine precision and recall for critical medical classification tasks.